

1 Log Questions

1. $\log_3(81) = 4$

2. $\log_5(\frac{1}{25}) = -2$

3. solve for x:

$$\log_2(x) = 5$$

$$x = 32$$

4. solve for b: $\log_b(8) = 3$

$$b = 2$$

5. $\log_b(1) = 0$ for every base b because $b^0 = 1$

6. At a glance $\log_2(9)$ can't be true, because no even number can't be multiplied by itself to become odd

7. $\log_2(0.5) = -1$

2 Log Rules

2.1 Multiplication

If $\log_b(a) = m$ and $\log_b(c) = n$, what is the $\log_b(a \times c)$?

$$a = b^m \quad \text{and} \quad c = b^n$$

$$a \times c = b^m \times b^n = b^{m+n}$$

$$\log_b(a \times c) = m + n$$

2.2 Division

If $\log_b(a) = m$ and $\log_b(c) = n$, what is the $\log_b(a \div c)$?

$$a = b^m \quad \text{and} \quad c = b^n$$

$$a \div c = b^m \div b^n = b^{m-n}$$

$$\log_b(a \div c) = m - n$$

2.3 Exponentiation

If $\log_b(a) = m$, what is $\log_b(a^k)$?

$$\begin{aligned}a &= b^m \\ \log_b(a^k) &= \log_b((b^m)^k) \\ &= \log_b(b^{km}) = km && \text{(sub m for } \log_b(a) \text{ for } a) \\ &= k \log_b(a)\end{aligned}$$

3 Log Rules Questions

1.

$$\begin{aligned}\text{a) } \log_2(8 \times 16) &= \log_2(128) \\ &= 7\end{aligned}$$

$$\begin{aligned}\text{b) } \log_2(8 \times 16) &= \log_2(8) + \log_2(16) \\ &= 3 + 4 = 7\end{aligned}$$

$$\text{2. } \log_3(27 \times 81) = \log_3(27) + \log_3(81) = 3 + 4 = 7$$

$$\text{3. } \log_2(8^5) = 5 \log_2(8) = 5 \times 3 = 15$$

$$\text{4. } \log_5(25^3) = 3 \log_5(25) = 3 \times 2 = 6$$

$$\text{5. compress and evaluate: } \log_2(8) + \log_2(4) = \log_2(8 \times 4) = \log_2(32) = 5$$

$$\text{6. } \log_b(x^2 y^3) = \log_b(x^2) + \log_b(y^3) = 2 \log_b(x) + 3 \log_b(y)$$

$$\text{7. a) } \log_2(8 + 8) = \log_2(16) = 4$$

$$\text{b) } \log_2(8) + \log_2(8) = 3 + 3 = 6$$

$$\text{Therefore: } \log_2(8 + 8) \neq \log_2(8) + \log_2(8)$$

8. The rules for logs are derived from the laws for exponents and there are no laws for sums.